

Line Tracking

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1. Experiment Purpose

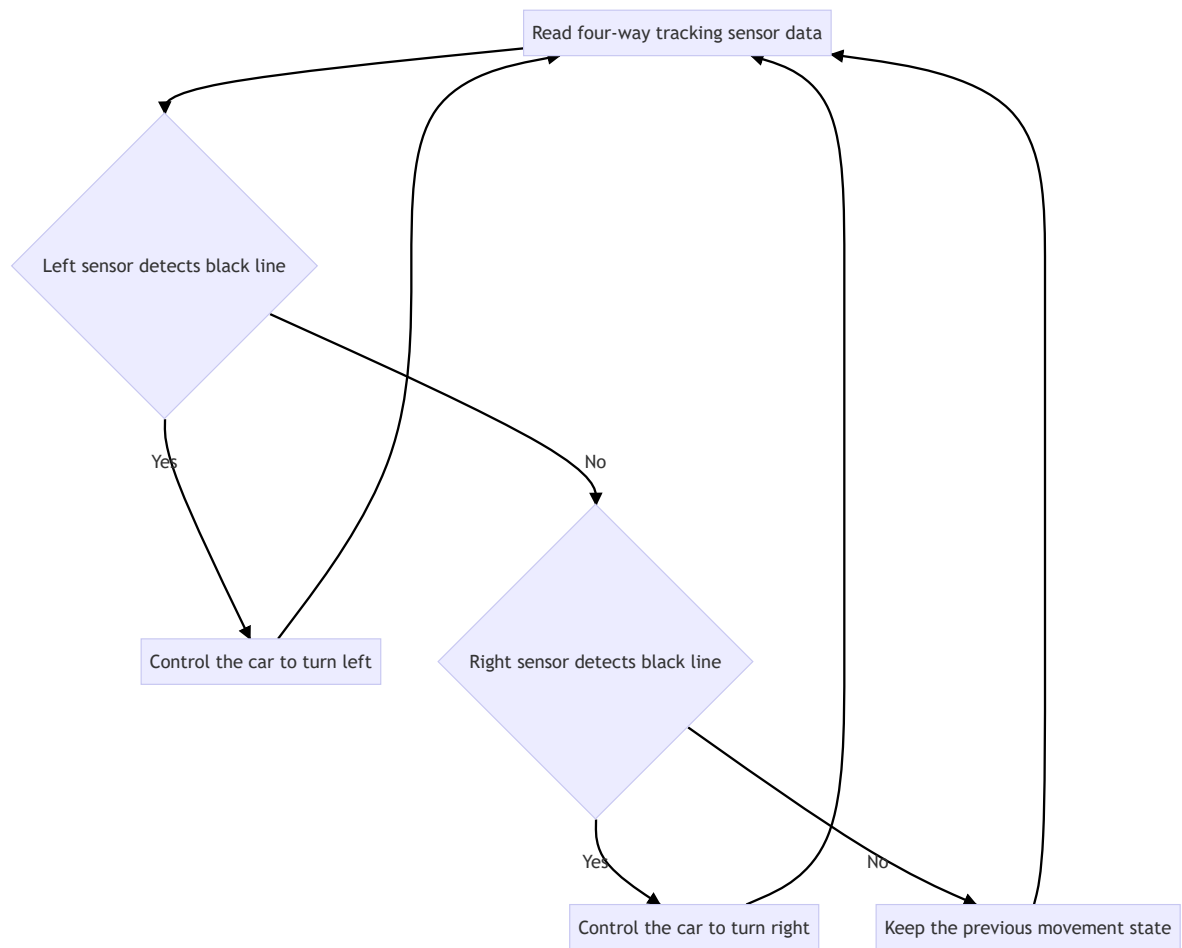
Earlier we learned how to read the data from the four-way tracking sensor. In this section, we will learn how to use this data to implement the line-following function.

2. Hardware Requirements

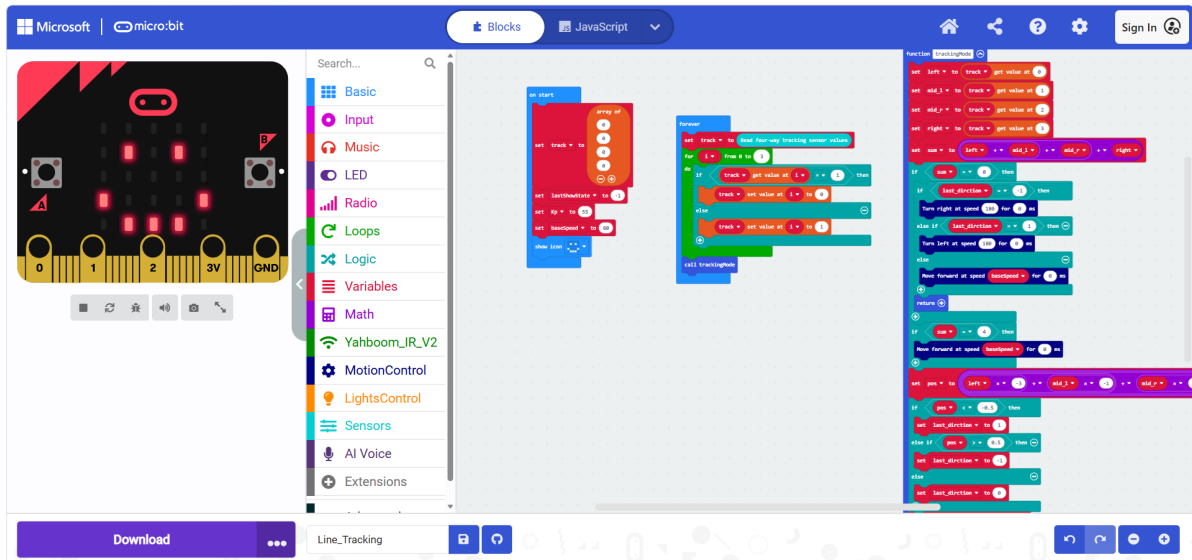
- Make sure the battery is fully charged and the four-way tracking sensor has been adjusted to a suitable sensitivity with the potentiometer.
- Lay the track flat on the ground.

3. Implementation Principle

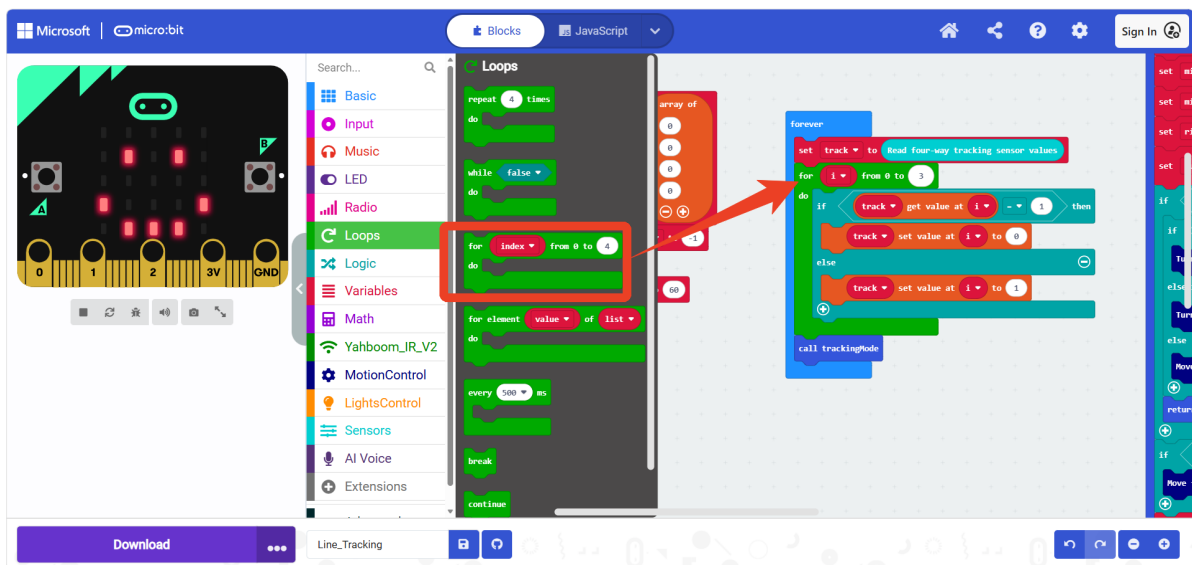
Read the data from the four-way tracking sensor, determine the state of the car's position on the line based on the data, and apply different controls for different states.



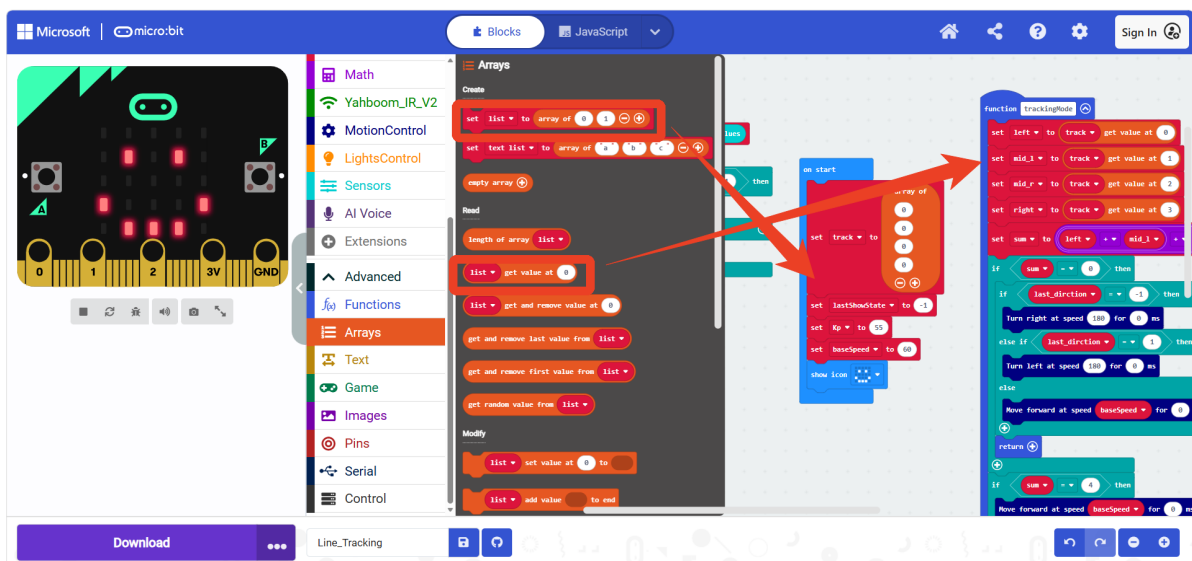
The program is as follows.



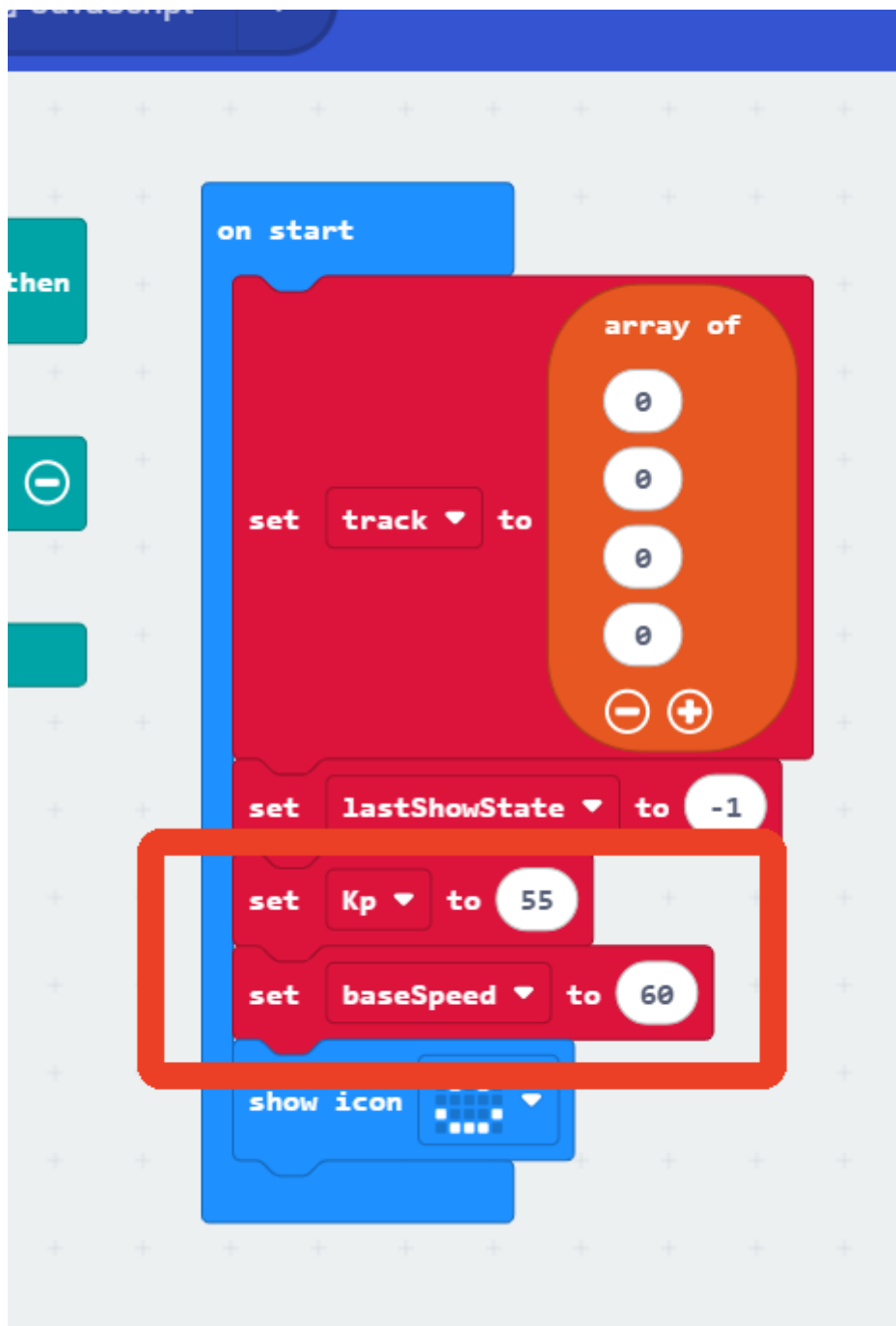
【For i from 0 to 3】 is located in the **Loops** category.



【Set () to ()】 is located in the **Arrays** category.



You can adjust the following parameters to set the car's driving speed and turning amplitude.



4. Experiment Steps

1. Before flashing, turn off the car switch (switch it to OFF), and connect the Micro USB cable between the computer and the Micro:bit. Note: connect to the Micro:bit board interface, not the Charging interface on the expansion board.
2. Flash `1.Line_Tracking.hex` to the Micro:bit. For the flashing method, refer to section "4. Flashing Programs" in "2. Introduction to Graphic Programming"
3. After flashing, unplug the data cable and place the car on the track to be followed.
4. Turn on the car switch (switch it to ON).

5. Experiment Phenomenon

Place the car on the track, and the car will follow the track to do line-following.

